Chapter 1: Motion

Speed is the distance travelled per unit of time. It is a scalar quantity, so it has a size but no direction. A car that covers 120 kilometres in 2 hours has an average speed of 60 kilometres per hour.

Velocity is speed in a given direction. Because velocity has a direction, it is a vector quantity. Two cars with the same speed can have different velocities if they travel in different directions.

Acceleration is the rate at which velocity changes with time. The formula is a = (v - u) / t, where u is the initial velocity, v is the final velocity and t is the time taken.

A negative acceleration is called deceleration. For example, a train slowing from 30 metres per second to 10 metres per second in 5 seconds has an acceleration of minus 4 metres per second squared.

Chapter 2: Forces

A force is a push or a pull. Newton's first law says that an object stays at rest or moves at constant velocity unless a net force acts on it. This property is called inertia.

Newton's second law states that the net force on an object equals its mass multiplied by its acceleration, F = m a. A 2 kilogram trolley accelerating at 3 metres per second squared needs a net force of 6 newtons.

Momentum is the product of mass and velocity, written p = m v. It is measured in kilogram metres per second.

In a closed system the total momentum before a collision equals the total momentum after it. This is the law of conservation of momentum. Newton's third law says that every action has an equal and opposite reaction.

Chapter 3: Energy

Work is done when a force moves an object. Work equals force multiplied by the distance moved in the direction of the force, W = F d. Work is measured in joules.

Kinetic energy is the energy of motion. It is given by KE = half m v squared. Doubling the speed of an object makes its kinetic energy four times larger.

Chapter 4: Waves

A wave transfers energy without transferring matter. The wavelength is the distance between two successive crests. The frequency is the number of waves passing a point each second, measured in hertz.

The speed of a wave is the product of its frequency and its wavelength, v = f x wavelength. A wave with frequency 50 hertz and wavelength 2 metres travels at 100 metres per second.